

Secure a Linux Box through the Right File Permissions



This article discusses how a Linux machine can be controlled by assigning permissions for accessing files and directories to owner users, group members and other users. Advanced techniques for user and group management are also covered.

Linux plays a dominant role in server management, worldwide, due to its powerful kernel, based on which a wide range of operating system flavours are available—Red Hat, OpenSUSE, CentOS and Ubuntu are a few of the popular ones. The high level of reliability, safety, security and flexibility provided by Linux has also made it popular among desktop users and mobile phone users (Android is built on Linux).

In an earlier article that I wrote for the December 2014 issue of *OSFY*, I argued that the security of Linux based OSs is in the hands of systems administrators. The topics of users and group management were discussed in detail. Further to adding and managing users, permission management is a necessary skill. Well controlled permissions and privileges implemented by a good administrator make a Linux box more secure; otherwise, it can remain vulnerable to numerous attacks and digital theft. So awareness of various security features provided by Linux and their implementation is necessary. In this article, I have focussed on permissions, their management and their effects, so that readers can secure their Linux machines better.

Permissions

Permissions mean conferring the authority to access data to an unauthorised person, by default. In the digital world,

primarily four kinds of permissions affect our interaction with computing machines: *read*, *write*, *execute* and *No permission*.

Read (r) allows a user to view the contents of a file, *write (w)* allows a user to modify or delete the contents, *execute (x)* allows a user to run a file such that the computer provides the relevant output after running the code provided, and *No permission (-)* restricts user access for a specified operation.

Permissions can be implemented on both files and directories. In the case of files, *read* allows the viewing of the contents, *write* allows the modification or deletion of the contents and *execute* allows the running of commands written within the file - this is in the case of shell scripts and other scripts. In case of directories, *read* allows the listing of the contents, *write* allows the creation or deletion of files and directories within a directory, and *execute* allows the opening of a directory. Without *execute* permissions, a directory cannot be read.

To view the permissions of particular contents, execute the following command in any directory:

```
ls -l
```

You will get the result shown in Figure 1.

A thorough exploration of users and groups was done in my earlier article. In the snapshot shown Figure 1, the